

Solving more complex quadratic equations

Quadratic equations do not always appear in the standard form of $ax^2 + bx + c = 0$. Instead, several x^2 terms, x terms, and numbers may appear on both sides of the equals sign. However, if all terms appear at least once, the equation can be rearranged in the standard form, and solved using the same methods as for simple equations.

This equation is not written in standard quadratic form, but contains an x^2 term and a term multiplied by x so it is known to be one. In order to solve it needs to be rearranged to equal 0.

Start by moving the numerical term from the right-hand side of the equals sign to the left by adding its opposite to both sides of the equation. In this case, -7 is moved by adding 7 to both sides.

Next, move the term multiplied by x to the left of the equals sign by adding its opposite to both sides of the equation. In this case, $2x$ is moved by subtracting $2x$ from both sides.

It is now possible to solve the equation by factorizing. Draw a table for the possible numerical values of x . In one column, list all values that multiply together to give the c term, 20; in the other, add them together to see if they give the b term (9).

Write the correct pair of factors into parentheses and set them equal to 0. The two factors of the quadratic ($x + 5$) and ($x + 4$) multiply together to give 0, therefore one of the factors must be equal to 0.

Solve the quadratic equation by solving each of the bracketed expressions separately. Make each bracketed expression equal to 0, then find its solution. The two resulting values are the two solutions to the quadratic equation: -5 and -4 .

$$x^2 + 11x + 13 = 2x - 7$$

these terms need to be moved to other side of equation for it to equal 0

7 has been added to this side ($13 + 7 = 20$)

$$x^2 + 11x + 20 = 2x$$

7 has been added to this side, which cancels out -7 , leaving $2x$ on its own

adding $-2x$ to $11x$ gives $9x$

$$x^2 + 9x + 20 = 0$$

subtracting $2x$ from this side cancels out $2x$

list possible factors of $c = 20$

add the factors to find their sum

Factors of +20	Sum of factors
20, 1	21
2, 10	12
5, 4	9

all sets of numbers in this column multiply to give 20

stop when the factors add to the b term, 9

parentheses set next to each other are multiplied together

$$(x + 5)(x + 4) = 0$$

entire equation equals 0

solve for first value

$$x + 5 = 0$$



$$x = -5$$

subtract 5 from both sides to isolate x

one possible solution is -5

solve for second value

$$x + 4 = 0$$

subtract 4 from both sides to isolate x

$$x = -4$$

another possible solution is -4

LOOKING CLOSER

Not all quadratic equations can be factorized

Some quadratic equations cannot be factorized, as the sum of the factors of the purely numerical component (c term) does not equal the term multiplied by x (b term). These equations must be solved by formula (see pp.192–193).

$$x^2 + 3x + 1 = 0$$

b term (3)

c term (1)

both sets of numbers multiply together to give c (1)

Factors of +1	Sum of factors
1, 1	2
-1, -1	-2

a sum of $+3$ is needed as the b term is 3

The equation above is a typical quadratic equation, but cannot be solved by factorizing.

Listing all the possible factors and their sums in a table shows that there is no set of factors that add to b (3), and multiply to give c (1).